



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO GOPALAKRISHNA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

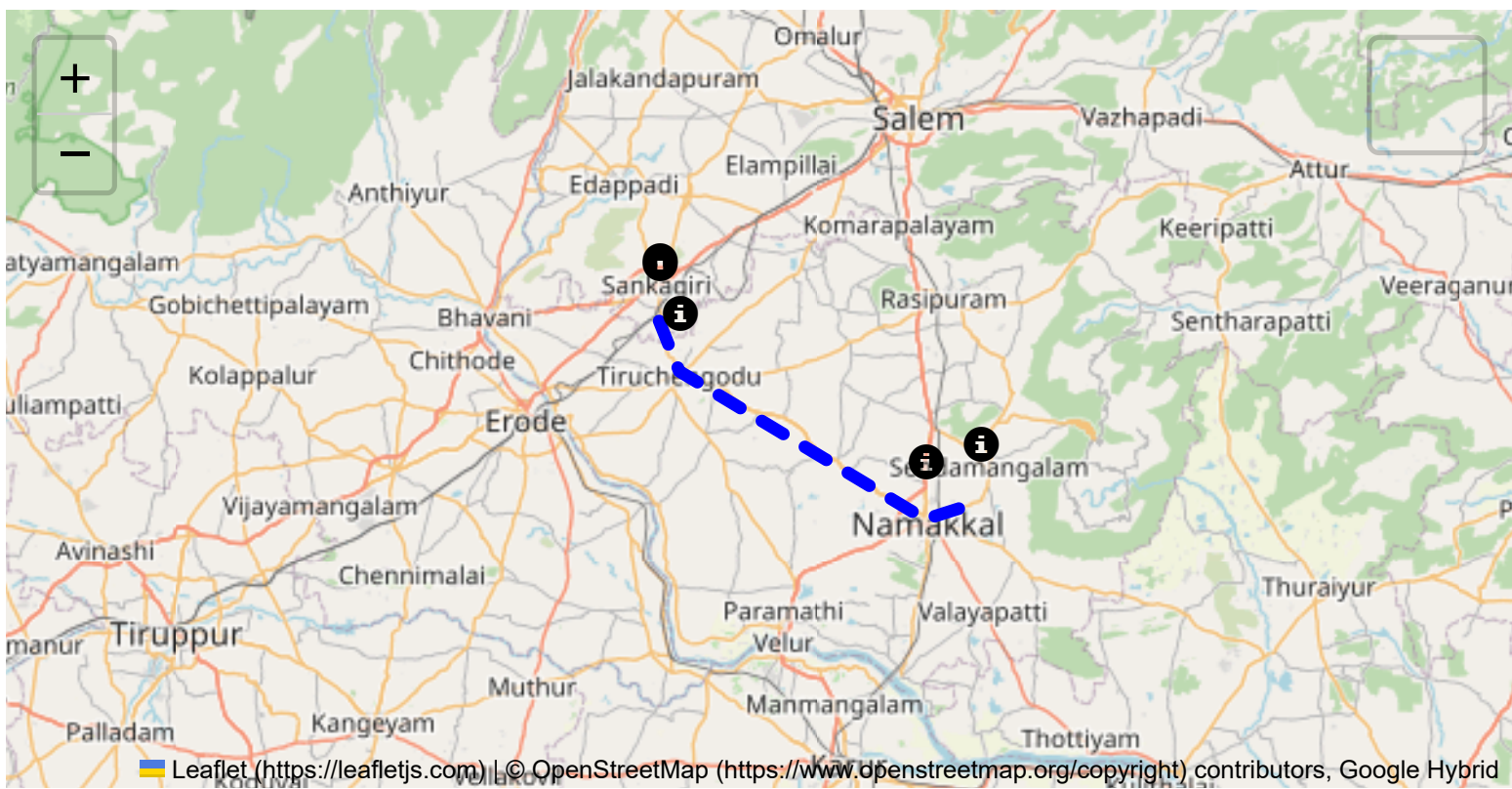
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

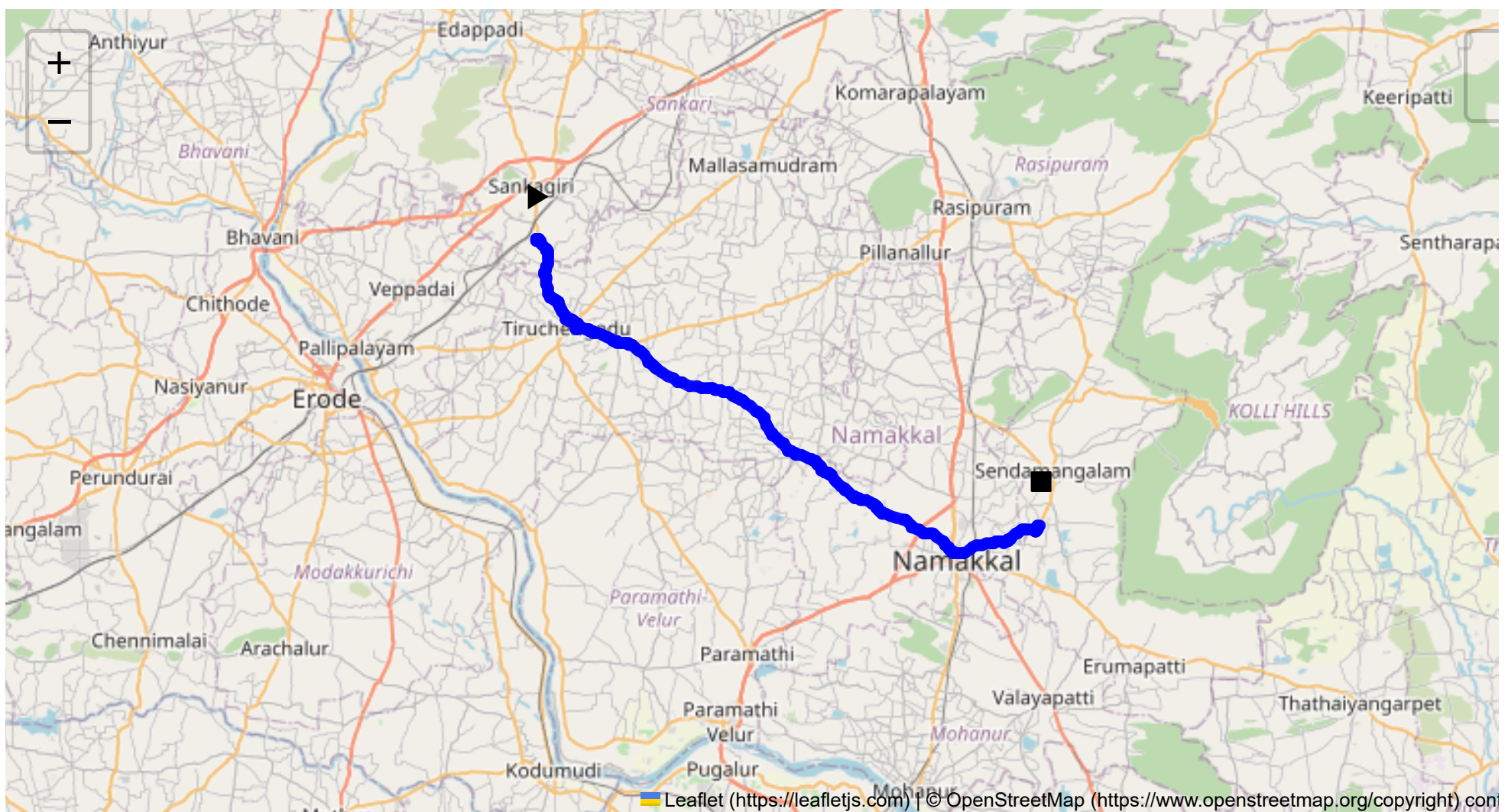
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 50.99 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.243513, 78.224785)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route begins in Sangagiri and proceeds through Puthur, Thondikaradu, and Thillaipuram before reaching Kondamanayakkanpatti. This route covers mixed rural and semi-urban areas with a combination of state highways and district roads.

2. Typical Weather Conditions and Potential Weather-related Hazards:

- **Monsoon Season (June-September):** Heavy rains can lead to waterlogged roads and decreased visibility.
- **Summer (March-May):** High temperatures can cause discomfort and heat-induced vehicle issues.
- **Winter (December-February):** Generally mild with occasional fog that may reduce visibility in the early mornings.

3. Analysis of Traffic Patterns:

- **Peak Hours:** Mornings (7-9 AM) and evenings (5-7 PM) through town centers like Namakkal and Tiruchengode.
- **Congestion-prone Areas:** Market regions in Namakkal and Tiruchengode, especially near schools and municipal areas.

4. Assessment of Road Quality and Infrastructure:

- **Sangagiri to Tiruchengode:** Roads are generally well-maintained but can have potholes and worn markings.
- **Tiruchengode to Namakkal:** Better road conditions with wider lanes.
- **Namakkal to Kondamanayakkanpatti:** Rural roads might be narrower and less maintained.

5. Suggestions for Alternative Routes for Emergencies:

- In the event of major congestion or roadblocks, consider using the Namakkal by-pass to avoid city center traffic.
- For detours, local farm roads can be an option, but they're typically suited for lighter vehicles.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- **Restrictions in time:** Transport of hazardous materials may be restricted during peak hours within urban areas.
- **Specific permits** may be required for transporting certain types of hazardous materials through populated areas.

7. Overview of Historical Incidents:

- Historical data suggests occasional incidents involving heavy vehicles, primarily due to navigational errors or vehicle overturns in congested areas.
- No recent major incidents involving hazardous materials specifically, but caution is advised near schools and crowded town centers.

8. Environmental Considerations and Sensitive Areas:

- Several schools such as the Girls High School in Thondikaradu are along or near the route, demanding cautious driving.
- Agricultural lands on either side imply sensitive environmental areas; any spillage could impact crops.

9. Analysis of Communication Coverage:

- Generally good coverage in urban areas such as Namakkal and Tiruchengode.
- Possible dead zones in rural sections between towns should be identified ahead of travel.

10. Estimated Emergency Response Times:

- **Urban Areas (Namakkal, Tiruchengode):** Estimated at 15-30 minutes due to traffic density.
- **Rural Segments:** 30-45 minutes due to distance from major centers and potentially difficult terrain.

11. Overall Summary of Risk Assessment:

- **Risks Identified:** Potential for traffic congestion, environmental sensitivity near agricultural land, variable road conditions, and communication dead zones.
- **Mitigation Measures:** Plan travel during off-peak hours, ensure vehicle maintenance to withstand weather changes, and maintain up-to-date permits and knowledge of local regulations.
- **Conclusion:** The route involves moderate risk, requiring careful planning and awareness of both environmental and traffic conditions. With preparations, risks can be adequately managed for safe transportation.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	High	11.38190, 77.90254	15 KM/Hr
7	Turn	Medium	11.38182, 77.90259	30 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	Medium	11.22481, 78.16539	30 KM/Hr
10	Turn	Medium	11.22408, 78.17204	30 KM/Hr
11	Turn	Medium	11.23996, 78.21411	30 KM/Hr
12	Blind Spot	Blind Spot	11.23912, 78.22323	10 KM/Hr
13	Turn	High	11.24010, 78.22322	15 KM/Hr
14	Turn	Medium	11.24040, 78.22355	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
16	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
17	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
18	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
19	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
20	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
21	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
22	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
26	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
27	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
28	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
29	clinic	Dhanvanthiri Physiotherapy	11.2245256, 78.1638528	30 km/h	Medium
30	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
31	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
33	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
34	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
35	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
36	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
37	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
38	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
39	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
40	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
23	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
24	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
32	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38182, 77.90259



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22481, 78.16539



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22408, 78.17204



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23996, 78.21411
